

Toolchain support matrix

Baseline: GCC 14 with libstdc++ 14, and Clang 18 with libc++ 18 (the repo's CMake adds -stdlib=libc++ for Clang; see cmake/warnings.cmake). Verified by building this repo. Update as sessions add features.

Feature	GCC 14 / libstdc++	Clang 18 / libc++	Notes and fallback
std::expected	OK	OK	Clang 18 with libstdc++ has none: libstdc++ gates it on __cpp_concepts >= 202002L, which Clang reports only from 19. This is why Clang builds use libc++.
std::print / println	OK	OK	
Range formatting (std::println("{ }", vec))	MISSING (GCC 15)	OK	demos/s02/format_ranges.cpp gated on __cpp_lib_format_ranges.
std::move_only_function, std::out_ptr	OK	MISSING (out_ptr: libc++ 19; move_only_function: not yet)	demos/s02/tour_cpp23.cpp gated on the feature-test macros.
std::make_format_args	takes lvalues	takes lvalues	C++23 DR P2905: pass a named variable, not a literal.
std::format, std::formatter specializations	OK	OK	libc++ requires the formatter's format() member to be a template on the context type (it checks against a compile-time context).
from_chars for double	OK	MISSING (libc++ 20)	__cpp_lib_to_chars is undefined on libc++ 18; the Session 1 solution

			and demos/s01/from_c hars.cpp fall back to strtod under #ifndef __cpp_lib_to_chars.
Deducing this	OK	OK	Session 3 solution (SensorStats::add) , constexpr, chained on lvalues and rvalues.
operator<=,>, designated initializers, using enum, uz, auto(x), to_underlying, unreachable	OK	OK	Session 1 demos.
constexpr, constexpr std::array/optional/ string_view/span in constant expressions	OK	OK	Session 3 solution. std::as_bytes is not constexpr on either (reinterpret_cast).
Concepts, abbreviated function templates, requires-clauses, fold expressions	OK	OK	Session 3 solution.
[[assume]]	OK (GCC 13+)	MISSING (Clang 19)	Clang 18 warns on the unknown attribute; kept out of compiled demos.
Range-for temporaries lifetime (P2718)	GCC 15	Clang 19	Slide shows the C+ +20 init-statement workaround.
std::generator	OK (GCC 14)	MISSING (libc++: not yet)	Session 5 solution gates records() on __cpp_lib_generat or.
std::flat_map / flat_set	MISSING (GCC 15)	MISSING (libc++ 20 has flat_map; flat_set from libc++ 21)	demos/s02/ tour_flat_map.cpp gated on __cpp_lib_flat_map ; slide uses Compiler Explorer with a

std::mdspan	MISSING (GCC 15)	OK (libc++ 17+)	trunk compiler. Slide only; Compiler Explorer.
std::stacktrace	OK, link -lstdc++exp	MISSING	demos/s02/ tour_stacktrace.cpp gated on __cpp_lib_stacktrace; CMake adds the link library for GCC.
std::optional monadic ops, std::expected value_or/error_or	OK	OK	Session 2 solution. transform(&T::member) does not compile on either, temporary or named (the result would be optional<T&>); use a lambda.
std::string::contains / ends_with	OK	OK	Session 2 solution and tests.
Named modules	OK (GCC 14, #include before import)	OK (Clang 18)	demos/s05/ modules, -DCOURSE_MODULES=ON, Ninja generator, CMake 3.28+.
import std;	GCC 15	libc++ 17+ with CMake 3.30+	Not built here (CMake 3.28).
Parallel algorithms	OK; TBB needed for actual parallelism	-fexperimental-library (added by CMake)	demos/s04/ parallel.cpp.
jthread, stop_token, counting_semaphore	OK	OK with -fexperimental-library (libc++ 18; CMake adds it)	Session 5 solution; clean under ThreadSanitizer on both.
views::zip, chunk_by, split, take, filter, iota; ranges::to; ranges::contains; projections	OK	OK	Session 4 solution.
views::enumerate	OK	MISSING (libc++ 20)	Solution uses views::zip(views::iota(1uz), ...)

views::chunk, slide, OK
stride, adjacent,
zip_transform,
cartesian_product,
join_with;
fold_left_first,
fold_right;
ranges::iota
const views

MISSING (also on
Apple Clang 17 /
Xcode 26)

instead.
demos/s04/
{windows,zip_enu
merate,zip_family
}.cpp gated on the
__cpp_lib_ranges_*
macros; they print a
notice instead.

filter_view/
drop_while_view
are not const-
iterable on either
(by design); take
ranges by R&&.